

# Differentiable-by-Construction FEM for Freeform Open-Radiator Design: A Curved Conformal Antenna Structured-Grid Inverse Design Cannot Reach

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## Abstract

Differentiable electromagnetic (EM) simulators have made photonic inverse design routine, but the dominant tools optimize a permittivity *density* field on a structured Yee/Cartesian grid, a representation that staircases curved metal and is a poor fit for open radiators terminated by a perfectly matched layer (PML). We present a tensor-compiled, differentiable-by-construction 3-D Nédélec H(curl) finite-element method (FEM) that exposes an exact, single-factorization discrete *shape* (moving-boundary / node-motion) adjoint through the full open-radiator forward: a matched box-UPML tensor material, lossy complex- $\varepsilon$  FR-4, and a lumped feed port assembled into one complex-symmetric pencil  $A(\omega) = K(\nu) - \omega^2 M(\varepsilon) + (j\omega/Z_s)S_p$ . Forward-mode “Dual-twin” element kernels give exact  $\partial K/\partial X$ ,  $\partial M/\partial X$ ,  $\partial b/\partial X$ , so the transpose adjoint reuses the single forward factorization ( $n_{\text{fact}} = 1$  per band frequency). We drive a high-DOF harmonic mesh-morph parametrization (73 active boundary DOFs) to gradient-design a *freeform curved-metal conformal* radiator: the optimizer reaches a genuine target-hit terminal that drives the entire design band below  $-10$  dB return loss (worst-of-band  $-5.51 \rightarrow -12.06$  dB), with the band gradient FD-validated at relative error  $1.17 \times 10^{-10}$ , enforced passivity  $|S_{11}| \leq 1$ , and a non-inversion guard. We then *measure* why a structured-grid density method cannot reach the same design, on two independent axes committed to the repository: geometric fidelity (rasterizing the curved conductor onto a Yee grid leaves a perimeter error that does not vanish but plateaus at  $\sim +14\%$ ;  $\sim 1 \mu\text{m}$  boundary fidelity would need  $\sim 250$  cells/mm) and compute (measured Meep scaling gives  $\sim R^3$  cells and  $\sim R^4$  wall-clock, so a faithful discretization is computationally intractable for single-process, single-node Meep 1.34.0 on a 61 GB host — a RAM wall by  $R \geq 14$  on that 61 GB box,  $\sim 10^{15}$  cells at the curve-faithful resolution, with the higher-resolution and multi-node cost stated as a labeled projection). We did not run a converged FDTD-density optimization; the intractability is shown by measured scaling and projection, and such a run is left as future work. We position the narrow, honestly-scoped novelty against prior EM shape adjoints and the structured-grid density ecosystem.

## 1 Introduction

Inverse design has become the default workflow for nanophotonic components: given a differentiable forward Maxwell solver, one back-propagates a figure-of-merit to a large parameter vector and lets gradient descent discover non-intuitive structures. The canonical “killer” demonstrations — the compact wavelength demultiplexer of Piggott et al. [Piggott et al. \(2015\)](#) and the high-NA achromatic metalens of Chung and Miller [Chung and Miller \(2020\)](#) — are guided-wave or periodic *photonic* problems, and the workhorse tools (CEVICHE [Hughes et al. \(2019\)](#), Meep’s adjoint solver [Hammond](#)

et al. (2022), SPINS-B Su et al. (2019), FDTD Mahlau et al. (2024, 2026), invrs-gym Schubert (2024)) optimize a permittivity *density* field discretized on a structured Yee/Cartesian grid. This representation is a superb fit for topology optimization of dielectric photonics, but it has two structural mismatches with a large and practically important class of RF/antenna problems: (i) it *staircases* curved metal, so a conformal radiator wrapped around a cylinder is only ever approximated at a density-limited grid cost; and (ii) it optimizes “where is there material” rather than “where is the metal boundary,” which is the natural design variable for a driven, PML-terminated *open radiator*.

We take the differentiable-simulation lens to that underserved corner. The substrate is a tensor-compiled, differentiable-by-construction 3-D Nédélec H(curl) FEM on *unstructured* tetrahedral meshes; the payload is a moving-boundary shape-adjoint geometry class that treats the positions of the conductor boundary nodes as the design variables and moves them — the metal boundary is represented exactly, at every step, on a body-fitted mesh. The headline is a design result: end-to-end gradient design of a freeform curved-metal conformal antenna that drives the entire design band below  $-10$  dB return loss. We then close the comparative loop the title asserts by *measuring*, on two committed axes, that a structured-grid density method cannot practically reach the same design at the resolution the curved geometry demands (Section 5).

We are deliberately narrow about the novelty. Node-motion EM shape adjoints on unstructured meshes already exist (Section 2); this is explicitly *not* a claim of “first shape adjoint in EM.” The contribution is a specific, previously-unoccupied *combination* that we state precisely, once, in Section 2.

Our contributions are:

- A differentiable-by-construction open-radiator FEM *shape* adjoint: forward-mode AD Dual-twin element kernels yield exact  $\partial K/\partial X$ ,  $\partial M/\partial X$ ,  $\partial b/\partial X$  through the radiating complex-symmetric pencil  $A(\omega) = K(\nu) - \omega^2 M(\varepsilon) + (j\omega/Z_s)S_p$ , so the transpose adjoint reuses the *single* forward factorization per band frequency ( $n_{\text{fact}} = 1$ ), with enforced passivity  $|S_{11}| \leq 1$  and a non-inversion / mesh-distortion guard (Section 3).
- A high-DOF freeform boundary parametrization (harmonic / P1-Laplace mesh-morph, 73 active boundary DOFs) that generalizes single-parameter tuning to freeform shape while keeping the volumetric mesh valid, FD-validated to relative error  $\sim 10^{-10}$  (Section 3).
- A headline result: end-to-end gradient design of a curved-metal conformal radiator that drives the entire band below  $-10$  dB return loss on a geometry a Yee-grid density method staircases, with the worst-of-band return loss reported in full in Section 4.
- An honest positioning of the narrow white space against prior EM shape-adjoint and differentiable-FEM work, plus a *measured* two-axis evaluation — geometric fidelity and compute — showing that structured-grid density FDTD is not merely less accurate but computationally intractable on this curved radiator at the resolution its geometry demands (Sections 2, 5).

## 2 Related Work

**Structured-grid density-based differentiable EM (the contrast class).** The dominant differentiable-EM tools optimize a permittivity/density field on a Yee/Cartesian grid: Meep’s adjoint solver Hammond et al. (2022) (built on the Meep FDTD engine Oskooi et al. (2010)), the forward-mode-differentiated CEVICHE Hughes et al. (2019) and its challenge suite, SPINS-B Su et al. (2019), the JAX-based 3-D FDTD framework FDTD Mahlau et al. (2024, 2026), and the invrs-gym benchmark toolkit Schubert (2024). These methods perform topology optimization of a

continuous density that is later binarized — a recipe applied to conductor-based antenna design as early as Erentok and Sigmund’s sub-wavelength antenna work [Erentok and Sigmund \(2011\)](#), which likewise optimizes material occupancy on a fixed grid rather than the position of a curved metal boundary. Their canonical successes — the Piggott et al. WDM demultiplexer [Piggott et al. \(2015\)](#) and the Chung–Miller achromatic metalens [Chung and Miller \(2020\)](#) — are guided or periodic photonics, not open radiators terminated by PML.

**The current structured-grid state of the art is better than naive staircasing — and still does not reach this problem.** Our contrast class must be scoped against the best current structured-grid practice, not a strawman. Subpixel-smoothed projection (SSP) [Hammond et al. \(2025\)](#) and its twice-differentiable successor SSP2 [Romano et al. \(2026\)](#) give the Meep density adjoint differentiable, subpixel-accurate *binarized* boundaries on the structured grid — for the material classes that adjoint supports. Commercially, Tidy3D’s autograd [Flexcompute, Inc. \(2026\)](#) spans polyslab-vertex, dispersive-material, and (on its development branch) triangle-mesh vertex gradients, with conformal-PEC subpixel averaging in the forward solver and a grid-aligned rectangular patch-antenna adjoint example. What none of these provides is the case this paper targets: as of this writing, no structured-grid AD framework provides shape derivatives of a curved *metallic* radiator through a driven-port objective — Meep’s material-grid adjoint excludes dispersive and metallic media [Meep developers \(2026\)](#), so the SSP/SSP2 subpixel boundaries do not extend to the conductor itself; FDTD’s time-reversal AD requires lossless propagation [Mahlau et al. \(2026\)](#); and Tidy3D’s autograd (2.12-dev) stops at gradients of dielectric embedded *in* PEC, not of the metal boundary itself [Flexcompute, Inc. \(2026\)](#), and is closed-source and cloud-executed besides. The title’s “cannot reach” is therefore a claim about curved-metal boundary fidelity under a driven-port objective in current structured-grid AD frameworks — not a wall-clock claim (the compute axis of Section 5 is separately scoped there).

**AD shape optimization still on a rectilinear grid.** Hooten et al. [Hooten et al. \(2025\)](#) accelerate shape optimization with automatic differentiation, but map shape parameters to a permittivity distribution *painted on a fixed rectilinear grid*. This reinforces, rather than closes, the white space we target: the design variable is boundary shape, but the discretization remains structured and the metal boundary is still resolved by a density field rather than a body-fitted mesh. Liu and Bonar [Liu and Bonar \(2024\)](#) likewise obtain subpixel-smoothed rasterized shape gradients through an image representation on a fixed grid, again for dielectric photonics rather than driven metal radiators.

**True EM shape (node-motion) adjoints on unstructured meshes exist.** We are explicit that node-motion EM shape adjoints are not new. Wang and Anderson [Wang and Anderson \(2011\)](#) present a discrete Maxwell shape adjoint on unstructured meshes, but in 2-D, time-domain, lossless, PEC-walled, with *no PML, no ports, and no S-parameters*. Ghassemi et al. [Ghassemi et al. \(2013\)](#) evolve a microstrip *antenna* geometry via an FEM adjoint sensitivity, but with a handful of *low-DOF control vertices* and a *hand-derived* adjoint. Ham, Mitusch, and Schmidt [Ham et al. \(2020\)](#) provide *automatic* FEM shape derivatives (Firedrake pyadjoint) for transient PDEs in general, but not the assembled open-radiator EM forward. Outside FEM, boundary- and integral-equation methods carry the newest PEC shape-sensitivity work: Takahashi [Takahashi \(2025\)](#) derives time-domain boundary-integral shape derivatives (forward and adjoint) for PEC scatterers, Arens et al. [Arens et al. \(2026\)](#) shape-optimize freeform tubular PEC scatterers via EFIE domain derivatives, and Balouchev et al. [Balouchev et al. \(2024\)](#) isogeometrically shape-optimize multi-tapered coaxial

baluns — a driven RF component — with an integral-equation solver. These further bound any “first PEC shape sensitivity” reading (which we do not claim), while remaining scattering- or IE-based: no volumetric UPML + lumped-port assembly, and no AD-composable adjoint. On the unstructured time-domain side, DGTD metasurface inverse design still resorts to Bayesian rather than adjoint search Gelly et al. (2026), consistent with the gap this paper fills.

**Defensible white space, stated precisely.** No found work occupies this exact combination: a 3-D frequency-domain Nédélec H(curl) FEM shape adjoint through the *full* open-radiator forward (box-UPML tensor material + lumped ports + lossy complex- $\varepsilon$  + passive  $|S_{11}|$ ), differentiable-by-construction with single-factorization reuse, applied to a *freeform, high-DOF curved-metal conformal* design. The three stacked differentiators versus the prior art are: (1) freeform / high-DOF versus a few control points; (2) full open-radiator physics versus lossless PEC; and (3) differentiable-by-construction and composable versus a hand-derived adjoint. This is the single statement of the novelty; later sections refer back to it rather than restate it.

### 3 Method

#### 3.1 The open-radiator forward pencil

The forward problem is the driven, frequency-domain Maxwell curl-curl equation discretized with first-order Nédélec (edge) elements on an unstructured tetrahedral mesh. At natural frequency  $\omega$  the assembled linear system is the complex-symmetric pencil

$$A(\omega) \mathbf{x} = \left[ K(\nu) - \omega^2 M(\varepsilon) + \frac{j\omega}{Z_s} S_p \right] \mathbf{x} = \mathbf{b}, \quad (1)$$

where  $K(\nu)$  is the curl-curl stiffness with reluctivity  $\nu = \Lambda^{-1}$ ,  $M(\varepsilon)$  is the mass matrix with permittivity  $\varepsilon = \varepsilon_r \Lambda$ , and  $S_p$  is the lumped-port surface term with feed reference impedance  $Z_s$ . The radiating boundary is a *matched box-UPML*: an anisotropic tensor-material absorber whose per-tet real weights  $\Lambda$  enter both  $K$  and  $M$ . Losses in the FR-4 substrate enter as a complex  $\varepsilon_r$ , making the pencil genuinely complex-symmetric rather than merely real-shifted. All frequencies in this work are dimensionless *natural units* as recorded by the solver; we do not assign physical GHz values. (The solver stores the fixture geometry with nominal `_mm` / `_ohm` suffixes on its fields — `bend_radius_mm`, `pml_thick_mm`, `port_resistance_ohm`; these suffixes are naming scaffolding, and we read the values as the dimensionless natural units the physics uses.)

The scattering parameter is extracted from the port voltage and current,

$$S_{11} = \frac{Z - Z_0}{Z + Z_0}, \quad Z = \frac{V}{I}, \quad I = \frac{2V_{\text{inc}} - V}{R}, \quad Z_0 = R, \quad (2)$$

with port resistance  $R = 50$  (dimensionless). Passivity requires  $|S_{11}| \leq 1$ ; we assert this at every forward, gradient, and line-search evaluation as a physics tripwire.

#### 3.2 Dual-twin element kernels and the single-factorization shape adjoint

The design variables are the physical coordinates  $X$  of the conductor boundary nodes. The shape derivative of the objective therefore requires exact  $\partial K / \partial X$ ,  $\partial M / \partial X$ , and  $\partial b / \partial X$  — the element-kernel Jacobians with respect to node motion. We obtain these by *forward-mode automatic differentiation of the closed-form element kernels*: each entry of the Nédélec local curl-curl, mass, and current-RHS kernels is re-evaluated in first-order dual arithmetic  $r + d\epsilon$  ( $\epsilon^2 = 0$ ). Seeding one node coordinate’s tangent with  $d = 1$  (all others 0) returns, in the dual parts, the exact partials of every local matrix entry with respect to that coordinate. These “Dual-twin” kernels mirror the real

$f_{64}$  element kernels entry-for-entry (so their real parts reproduce the forward assembly), and are shared with the eigenvalue-sensitivity path that reuses the same element JVP. The tensor-material (box-UPML) twin carries the fixed real  $3 \times 3$  per-tet weights  $\Lambda$  as lifted constants (zero tangent), so the tangent flows only through geometry — the “fixed  $\Lambda$ ” convention of the box-UPML shape adjoint.

Because the pencil (1) is complex-symmetric, the transpose adjoint system uses the *same* operator as the forward solve. We therefore factor  $A(\omega)$  *once* per band frequency (a single complex LU) and reuse that factorization for both the forward solve and the adjoint solve:  $n_{\text{fact}} = 1$  per frequency. The nodal shape gradient at each band frequency is  $\lambda^\top (\partial A / \partial X) \mathbf{x}$  assembled from the Dual-twin element Jacobians, where  $\lambda$  is the adjoint solution.

### 3.3 High-DOF freeform boundary parametrization

To move from single-parameter tuning to freeform shape, we parametrize the boundary with a harmonic mesh-morph. Each free patch-conductor node is an independent radial-normal design DOF; the interior is extended by a P1-Laplace mesh-morph regularizer so the volumetric tetrahedra deform gracefully and stay non-inverted under large boundary deformation. The PML shell, the pinned feed port, the ground plane, and the outer boundary are held fixed (**fixed\_zero**). The multi-DOF design gradient is the per-column contraction  $\partial g / \partial X_p = \langle \text{grad}_{\text{node}}, D_p \rangle$  of the nodal adjoint gradient against the morph columns  $D_p$ . On the reported fixture this yields 73 active freeform boundary DOFs.

The band objective is the frequency-weighted sum of squared reflection magnitudes,

$$G(X) = \sum_f w_f |S_{11}(f; X)|^2, \quad (3)$$

with uniform band weights over  $\omega \in \{0.30, 0.35, 0.40\}$  (the v1 spec: impedance match + bandwidth only). The band nodal gradients are frequency-weighted, summed, and contracted through the morph columns in one shot.

### 3.4 Guards and validation

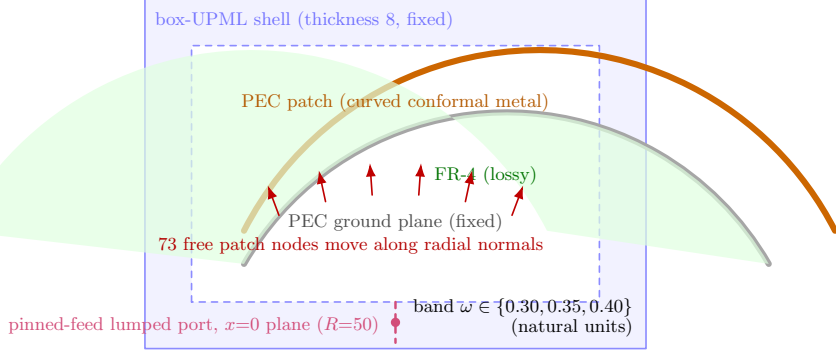
Two guards run at every evaluation. *Passivity*:  $|S_{11}| \leq 1$  across the whole band. *Non-inversion*: a per-tet signed-volume ratio  $|\det(X) / \det(0)| \geq \text{min\_vol\_ratio}$  (budget 0.25) rejects near-inverted tetrahedra produced by the morph. The full multi-DOF, multi-frequency band gradient is FD-validated before descent begins: the analytic directional derivative  $\nabla G \cdot \hat{d}$  is compared against a central finite difference of the *entire* port + UPML + lossy pipeline through the public `driven_solve_with_ports(MatchedU` path, on the curved mesh, along the steepest-descent direction at  $X = 0$ .

Figure 1 sketches the curved conformal radiator inside the box-UPML with the pinned feed port.

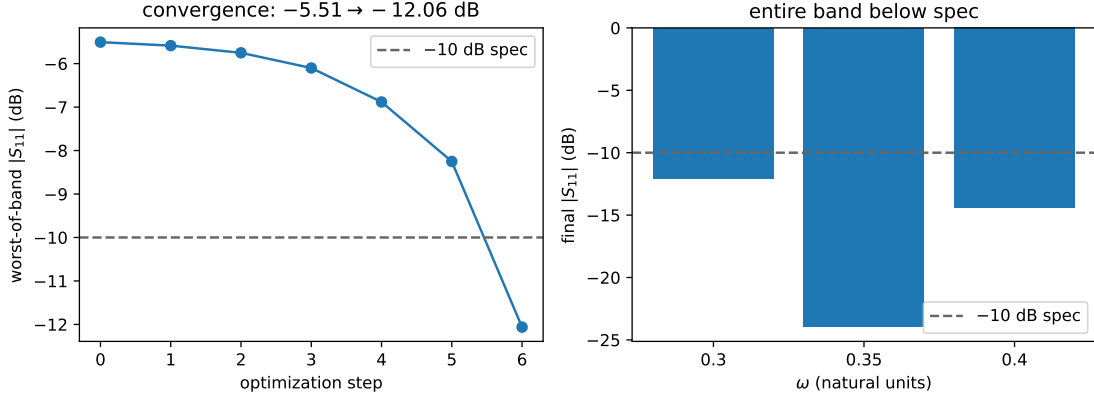
## 4 Results

**Fixture.** The reported design is the `bent_conformal` fixture: a flat FR-4 + PEC patch/ground slab wrapped around a cylinder of bend radius 40 (natural units) about the  $y$ -axis to produce genuinely curved conformal metal, with an 8 (natural units) box-UPML shell and an  $x = 0$  port plane. The discretization has 6173 edges, 997 nodes, and 4875 tetrahedra; the design vector has 73 active freeform boundary DOFs and a 50 (dimensionless) port resistance.

**Gradient correctness.** The full band gradient matches a central finite difference of the public driven pipeline on the curved mesh to relative error  $1.17 \times 10^{-10}$  (analytic and central directional derivatives both  $1.751288204 \times 10^{-1}$ , FD step  $10^{-6}$ , tolerance  $5 \times 10^{-3}$ ), with one factorization per band frequency.



**Figure 1:** Schematic of the freeform curved-metal conformal radiator: an FR-4 + PEC patch/ground slab wrapped around a cylinder of bend radius 40 (natural units, as recorded) inside a matched box-UPML shell, driven by a pinned-feed lumped port on the  $x = 0$  plane. The free patch-conductor nodes (73 active DOFs) move along their radial normals; the PML shell, port plane, ground plane, and outer boundary are held fixed.



**Figure 2:** Left: worst-of-band return loss versus optimization step ( $-5.51 \rightarrow -12.06$  dB over 6 accepted steps). Right: final per-frequency  $|S_{11}|$  (dB) at  $\omega \in \{0.30, 0.35, 0.40\}$  against the  $-10$  dB spec (dashed). All values from the committed artifact `conformal_results.toml`.

**Headline design result.** Bounded backtracking steepest descent (Armijo line search) on the 73-DOF design vector reaches a genuine `target_reached` terminal in 6 accepted steps (of a 600-step cap — a true target hit, not a step cap), with 0 total backtracks. The band objective (3) falls from  $1.897943102 \times 10^{-1}$  to  $3.411688781 \times 10^{-2}$ , a reduction factor of 5.56. The worst-of-band return loss improves from  $-5.51$  dB to  $-12.06$  dB, driving the *entire* design band below the  $-10$  dB spec. The final per-frequency reflection is

| $\omega$         | 0.30     | 0.35     | 0.40     |
|------------------|----------|----------|----------|
| $ S_{11} $ (dB)  | $-12.06$ | $-23.92$ | $-14.42$ |
| $ S_{11} $ (mag) | 0.2494   | 0.0637   | 0.1900   |

Figure 2 shows the convergence trajectory and the final per-frequency return loss against the  $-10$  dB target.

**Guards and determinism.** The non-inversion guard never bound: the worst per-tet signed-volume ratio was 0.572 against the 0.25 budget, at a maximum nodal displacement of 0.560

(natural units). Passivity  $|S_{11}| \leq 1$  held at every evaluation. The gradient norm fell from  $1.751 \times 10^{-1}$  to  $8.987 \times 10^{-2}$ . An independent fresh forward band solve at the final design (public `driven_solve_with_ports`, no adjoint) reproduces the optimizer’s final band objective to relative 0.0, and the run is bit-identical on re-run.

**Methodological foundation.** The composed driven SHAPE-adjoint stack — lossy complex- $\varepsilon$  adjoint, pinned/moving-feed lumped-port adjoint, box-UPML tensor-material adjoint — was validated by a prior single-DOF capstone (a separate, earlier artifact) that retuned a flat patch through the same FD gate. The conformal result extends that from 1 DOF to 73 freeform DOFs on curved geometry.

## 5 Evaluation: Structured-Grid Density FDTD is Intractable on the Curved Radiator

The title asserts a comparative claim — that a structured-grid density inverse design cannot reach this curved conformal radiator. We substantiate it not with one head-to-head optimization but with two independent *measured* axes, both committed to the repository (Section 7): geometric fidelity (Section 5.1) and compute (Section 5.2). **Honesty note.** We did *not* run a converged FDTD-density optimization of this geometry. The intractability is established by measured scaling laws plus clearly-labelled projections; a converged FDTD-density run would only re-confirm the intractability and is stated as future work (Section 6). Every number below traces to `benchmarks/fdtd_density_baseline/`.

### 5.1 Geometric-fidelity axis: the staircasing error does not vanish

The `bent_conformal` patch conductor is a circular arc (bend radius 40; the top and bottom conductor faces lie on radii 40.8 and 39.2, spanning a  $\phi_{\max} = 0.3$ -rad sweep, a  $\sim 24.1$ -wide feature). We rasterize this arc onto a Yee/Cartesian density grid at increasing resolution  $N$  and compare its perimeter, area, and boundary position against the exact arc (`staircasing_results.json`). The unstructured-tet reference is exact by construction: its nodes lie *on* the arc, so its boundary-position RMS is 0 to machine precision at *any* resolution and at fixed DOF.

| $N$ | cell size | perimeter rel. err. | area rel. err. | boundary RMS |
|-----|-----------|---------------------|----------------|--------------|
| 20  | 1.206     | 13.0%               | 13.58%         | 0.291        |
| 40  | 0.603     | 15.4%               | 0.33%          | 0.157        |
| 80  | 0.301     | 14.2%               | 0.33%          | 0.095        |
| 160 | 0.151     | 14.2%               | 0.15%          | 0.045        |

The area error converges (log-log slope  $\approx 1.96$  in  $h$ ) and the boundary position drifts down slowly (slope  $\approx 0.88$ ), but the *perimeter* error — the quantity that sets edge-current path length and thus resonance — does *not* converge to zero: it plateaus at a constant  $\sim +14\%$  (log-log slope  $-0.026$ , i.e. flat) as the grid is refined. A staircased boundary trades area accuracy for a persistent perimeter bias no amount of uniform refinement removes. To make the rasterized boundary as faithful as the conformal tet mesh (a  $\sim 1\ \mu\text{m}$  target) requires  $\sim 6029$  cells across the  $\sim 24.1$  feature —  $\sim 250$  cells/mm — which in 3-D is  $\sim 5.35 \times 10^4$  times the Yee cells of the finest grid tested. The conformal mesh reaches that fidelity at fixed DOF. Subpixel smoothing does not rescue the density method on this axis: SSP [Hammond et al. \(2025\)](#) and SSP2 [Romano et al. \(2026\)](#) give the Meep adjoint differentiable subpixel-accurate binarized boundaries for the material classes it supports, but the material-grid adjoint does not support dispersive or metallic media [Meep developers \(2026\)](#),

so the curved *conductor* boundary — the quantity measured here — remains staircase-limited in that stack.

## 5.2 Compute axis: measured $\sim R^3$ cells and $\sim R^4$ wall-clock

We measured the 3-D FDTD cost of the same box-UPML curved conformal open-radiator problem with Meep 1.34.0 [Oskooi et al. \(2010\)](#) (the vendored `reference/meep/docker/` container), single-process on an AWS `m6i.4xlarge` (16 vCPU Intel Ice Lake, 61 GB RAM), timing steady-state seconds/timestep over 61 steps after warmup (`meep_runtime_scaling.json`).

| $R$ (cells/mm) | Yee cells | s/step | peak RAM | dt     |
|----------------|-----------|--------|----------|--------|
| 4              | 10.3 M    | 0.233  | 1.6 GB   | 0.125  |
| 6              | 34.8 M    | 0.742  | 4.9 GB   | 0.0833 |
| 8              | 82.6 M    | 1.715  | 11.3 GB  | 0.0625 |
| 10             | 161.3 M   | 3.386  | 21.8 GB  | 0.05   |
| 12             | 278.7 M   | 5.761  | 37.4 GB  | 0.0417 |

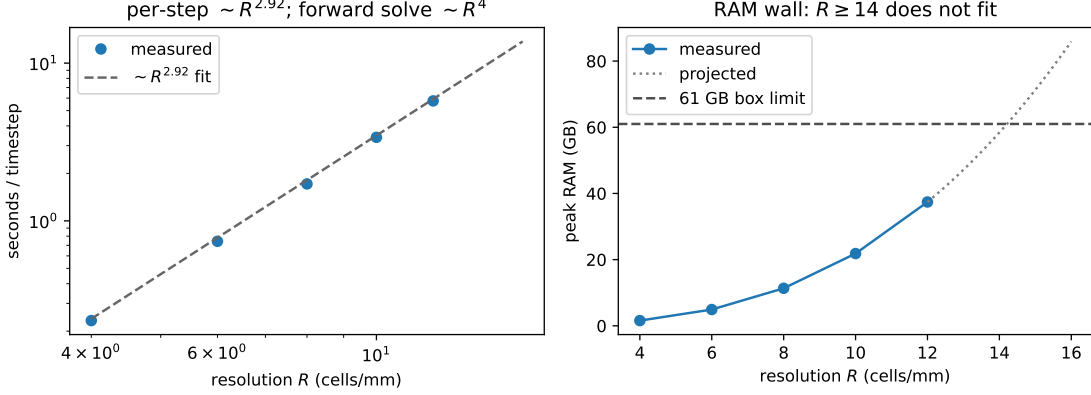
The cells count is exactly  $161280 R^3$  (structured Yee grid); the measured per-step time scales  $\sim R^{2.92}$  (memory-bandwidth-bound), peak RAM  $\sim R^{2.89}$ , and  $dt \propto 1/R$  (the CFL/Courant condition, measured  $0.125 \rightarrow 0.0417$ ). Because a fixed physical field-decay time then needs  $\sim R$  timesteps, a single forward solve scales  $\sim R^{2.92} \cdot R \approx R^{3.92}$ , i.e.  $\sim R^4$  (Figure 3).

**Measured absolute anchor.** On the same box, a single  $R = 8$  forward solve ran  $\geq 421$  timesteps and did *not* reach field decay within a 15-minute window — so  $\geq 12$  min per forward solve and  $\geq 24$  min per adjoint gradient, at a resolution that still badly staircases the curve ( $\sim 14\%$  perimeter error).

**Projections (extrapolated, not measured).** Extrapolating the measured power laws from the  $R = 8 \geq 421$ -step anchor (these are projections, clearly *not* measured runs): a 3-frequency,  $\geq 12$ -evaluation band optimization is  $\geq 14$  hours at  $R = 8$  and  $\geq 70$  hours at  $R = 12$  (the  $\sim R^{3.92}$  scaling of the  $R = 8$  lower bound). Peak RAM hits  $\sim 60$  GB at  $R = 14$  and  $\sim 83$  GB at  $R = 16$ , so  $R \geq 14$  does not fit on the 61 GB box — a hard RAM wall well before the curve is resolved. The resolution that would actually represent the arc faithfully,  $\sim 250$  cells/mm (Section 5.1), implies  $\sim 2.5 \times 10^{15}$  cells — roughly  $30\times$  beyond the RAM wall before any timestep cost is counted.

## 5.3 Head-to-head

The two axes combine into the comparative claim. On the geometric axis, a structured Yee grid cannot represent the curved conductor without a persistent  $\sim 14\%$  perimeter bias unless refined to  $\sim 250$  cells/mm; on the compute axis, that resolution — indeed any resolution past  $R \approx 14$  — is computationally intractable for single-process, single-node Meep 1.34.0 on a 61 GB host ( $\sim R^3$  cells,  $\sim R^4$  wall-clock, a RAM wall on that 61 GB box,  $\sim 10^{15}$  cells at curve-faithful resolution; the higher-resolution and multi-node cost is a labeled projection, and distributed or GPU FDTD would move the wall at added engineering cost). The present unstructured-tet Nédélec shape adjoint, by contrast, solved the full 73-DOF freeform curved-conformal band match to worst-of-band  $-12.06$  dB in 6 optimization steps, one reused factorization per band frequency, in seconds-to-minutes, exact at fixed DOF with no staircasing (Section 4). The conclusion the title asserts is therefore measured, not merely motivational: structured-grid density FDTD is not just less accurate on this curved open radiator — as measured for single-process, single-node Meep 1.34.0 on the 61 GB box, it cannot practically reach the design at the resolution the geometry demands.



**Figure 3:** Measured Meep 1.34.0 FDTD cost for the curved conformal open-radiator problem on AWS m6i.4xlarge. Left: seconds/timestep versus resolution  $R$  (cells/mm), with the fitted  $\sim R^{2.92}$  power law; a single forward solve adds a factor  $\sim R$  (CFL steps), giving  $\sim R^4$ . Right: peak RAM versus  $R$  with the 61 GB box limit marked —  $R \geq 14$  (projected) does not fit. Points are measured (`meep_runtime_scaling.json`); the shaded region is projected. Even the largest measured  $R = 12$  still staircases the conductor ( $\sim 14\%$  perimeter error).

**Scope of the intractability claim.** The compute axis is deliberately a *matched* comparison — matched conformal-boundary fidelity, single-process, open-source density-adjoint tooling (Meep 1.34.0) — and two prior-art objections deserve naming precisely because this scoping answers them. First, precomputed numerical-Green-function (PNGF) inverse design [Sun et al. \(2025\)](#) achieves near-real-time full-wave, lumped-port-driven antenna design — with fabricated 5G prototypes and reported speedups up to  $16,000\times$  over commercial solvers — via direct binary search over a pixelated design region; so “structured-grid antenna inverse design” is emphatically *not* intractable in general. PNGF is fast precisely because it forgoes both adjoint gradients and curved-boundary fidelity, so it does not reach the freeform curved-metal design class measured here. Second, hardware-accelerated FDTD such as Tidy3D [Flexcompute, Inc. \(2026\)](#) would move the wall-clock and RAM walls at added (closed-source, cloud-execution) cost — but the geometric-fidelity axis (Section 5.1) and the absence of metal-shape gradients (Section 2) are unchanged by faster hardware.

## 6 Discussion

The result substantiates a narrow but clean claim: a differentiable-by-construction open-radiator FEM *reaches* a freeform curved-conformal  $-10$  dB design by moving conductor-boundary nodes on a body-fitted mesh, with an exact, single-factorization shape adjoint and enforced physics guards; and a structured-grid density method cannot practically reach the same design at the resolution the curved geometry demands (Section 5). The design space is genuinely high-dimensional (73 independent DOFs; not one parametric knob), and the geometry is genuinely curved — the two properties that motivate a body-fitted node-motion adjoint over a structured density field.

**Limitations and threats to validity.** The scope of v1 is impedance match + bandwidth only; radiation pattern and gain (a near-to-far-field adjoint) are not addressed here. The GEODE design result is a single fixture at one bend radius and one band; we have not swept mesh resolution, bend radius, or band placement, so we make no claim about generality across radiator families — the comparative evidence in Section 5 is what lifts this from “one validated run” to a capability claim, but the reachability demonstration itself remains single-fixture and a sweep is warranted. On the

comparative side, we are careful about what is and is not measured: the two intractability axes (staircasing, runtime scaling) are *measured*; the full-optimization wall-clock figures ( $\geq 14$  h at  $R = 8$ ,  $\geq 70$  h at  $R = 12$ ) and the RAM wall at  $R \geq 14$  are *projections* extrapolating measured power laws from a measured  $\geq 421$ -step anchor — we did not run a converged FDTD-density optimization. We also note a reproducibility detail in the spirit of full disclosure: an initial optimization run mis-recorded a premature (step-cap) stop as a physics limit; a corrected line search reached the true `target_reached` terminal, which is the run reported here.

**Future work.** A converged FDTD-density optimization of this geometry — which would re-confirm the intractability rather than overturn it — and a low-DOF parametric baseline remain to be run. A near-to-far-field (NTFF) shape adjoint extending the objective from  $|S_{11}|$  to pattern/gain is future work and is not claimed here.

## 7 Conclusion

We presented a tensor-compiled, differentiable-by-construction 3-D Nédélec  $H(\text{curl})$  FEM that exposes an exact, single-factorization discrete shape adjoint through the full open-radiator forward — matched box-UPML tensor material, lumped ports, lossy complex- $\varepsilon$ , and a passive  $|S_{11}|$  — on unstructured tetrahedral meshes. Driving a 73-DOF harmonic freeform boundary morph, we gradient-designed a curved-metal conformal radiator that drives the entire design band below  $-10$  dB return loss (the per-frequency and worst-of-band figures are reported in full in Section 4), with the band gradient FD-validated to relative error  $1.17 \times 10^{-10}$ , enforced passivity, and a non-inversion guard. We then *measured*, on two committed axes, that a structured-grid density method cannot practically reach the same design: the staircasing perimeter error plateaus at  $\sim +14\%$ , and the FDTD cost scales as  $\sim R^3$  cells and  $\sim R^4$  wall-clock, hitting a RAM wall (measured for single-process, single-node Meep 1.34.0 on a 61 GB host; higher-resolution and multi-node cost stated as a labeled projection) well before the curve is resolved. The contribution is the narrow, previously-unoccupied combination of Section 2 — explicitly not a first EM shape adjoint. A converged FDTD-density optimization and a pattern/gain NTFF adjoint are the announced next steps.

## Artifact and code availability

All quantitative claims trace to artifacts committed on the `main` branch of the GEODE-FEM repository, publicly available at <https://github.com/rjwalters/geode-fem>. The headline design result is `benchmarks/patch_antenna_conformal/conformal_results.toml`. The evaluation-section measurements are in `benchmarks/fdtd_density_baseline/`: the geometric-fidelity axis in `staircasing_results.json` and the compute axis in `meep_runtime_scaling.json` (with `RUNTIME_SCALING.md`). The FDTD-density baseline itself is the vendored Meep 1.34.0 container under `reference/meep/docker/` (`conformal_baseline_3d.py`). The shape-adjoint implementation lives in `crates/geode-core/src/driven/shape`, `crates/geode-core/src/shape.rs`, `crates/geode-core/src/mesh/patch.rs`, and the runnable example `crates/geode-core/examples/patch_conformal_diffopt.rs`.

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